Pakistan J. Pharmacol. ISSN 0255-7088
Vol. 14(1), January 1997, pp.57-62

CHANGE IN NUCLEIC ACIDS (RNA AND DNA) CONTENTS, OF *MUSCA DOMESTICA*, UNDER THE EFFECTS OF NEEM COMPOUNDS AND DDVP

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ABSTRACT

Nucleic acids (RNA and DNA) contents were estimated in *Musca domestica* (PCSIR strain) after LD_{50} treatments of neem compounds (i.e. MargosanO™ , H-34, N6-b) and an organophosphate (DDVP). RNA and DNA contents were inhibited by all the undertest compounds. Treatment with Margosan-O™, H-34 and N6-b resulted in the inhibition of RNA content upto 23.37%, 23.55% and 33.32% respectively. Whereas the organophosphate, DDVP decreased the RNA content upto 9.67%. The effectiveness of the compounds for inhibiting RNA contents may be summarized as follows DDVP < Margosan-O™ < H-34 < N6-b. DNA contents also decreased by all the compounds. Inhibition was 19.49% by Margosan-O™ , 35.52% by H-34, 25.02% by N-6-b, and 28.82% by DDVP. The effectiveness of the compounds for inhibiting DNA contents may be summarized as follows. Margosan-O™ < N6-b < DDVP < H-34.

INTRODUCTION

Musca domestica L. is an important arbovirus vector throughout the world. This hygiene pest has a wide distribution over the globe. In Pakistan relative abundance is extremely high just after monsoon rains (August and September) and low in cool dry winter season. The control of *Musca domestica* L. using different synthetic, conventional pesticides is associated with tremendous problems like pollution and resistence etc These facts have diverted the attention of scientist towards natural products which are biodegradable, less persistent, safer for the environment and on the other hand the cost of natural products is lesser than conventional pesticides. Workers all over the world are busy with the investigations on natural products, especially neem compound with the idea of replacement of conventional pesticides by phytopesticides (Schmutterer, 1987).

To understand the physiological effects of neem compounds (Margosan-O™, H-34, RB-b, RB-a) and an organophosphate (DDVP) on nucleic acids present investigations were undertaken.

MATERIALS AND METHODS

For the estimation of Nucleic acids (RNA and DNA) batches of hundred adult flies each of similar age and size (3 day old) were taken and than treated with the LD_{50} value of the under test compound i.e. Margoson-O™ = 0.25 µg/fly, H-34 = 1.8 µg/fly, RB-b = 35 µg/fly, DDVP = 0.0086 µg/fly. Whereas a batch of 100 flies was kept as untreated control. After 24 hours of treatment, 50 alive insects weighting 300 mg were taken from each treated and untreated batch. RNA and DNA estimations were carried out according to Shakoori and Ahmed (1973), based on Schneider (1957).

RESULTS

Estimation of nucleic acids (RNA and DNA) was done after the treatments with LD_{50} values of the compound according to Shakoori and Ahmed (1973). It revealed a moderate inhibition of RNA and DNA by all the compounds. In DDVP treated batch the least inhibition was noted in RNA contents.

Margosan-O™ treated flies showed 23.37% inhibition in RNA contents and 19.49% inhibition in DNA. While in H-34 treated flies 23.55% inhibition in RNA and 35.52% inhibition in DNA contents was observed N6-b neem compound is crude in nature. The percentage of inhibition was 33.32% and 25.02% for RNA and DNA, respectively DDVP treated flies showed 9.67% decrease in RNA contents whereas 28.82% inhibition in DNA contents (Tables 1 and II).

DISCUSSION

Neem compounds show toxic and IGR effects. A large number of workers reported these properties of neem compounds (Chavan 1984, Naqvi, 1987, Khan *et al.*, 1991, Naqvi *et al.*, 1994, Naqvi *et al.*, 1995). Othaki and Williams (1970) reported that the insect body enzymes for the degradation of hormones like the molting hormones (MH), Scha and sehnal in (1973) reported that MH hormones activated the synthesis of RNA and JH simultaneously induced a duplication of DNA. This process causes such a severe disturbance in the insect body that it leads to death. It is because the RNA and DNA synthesis are mutually separated and exclude each other.

In the present findings all the test compounds inhibited RNA and DNA contents in adult *M. domestica*. Synthetic pesticide DDVP showed knockdown effect and inhibition of acetylcholinesterase. This compound inhibited lesser

Table I Comparison of the RNA content in *Musca domestica* L., (PCSIR strain) after LD_{50}'s treatment of the respective tested compounds

S.No.	Compound	LD_{50}'s µg/fly	RNA content Mean unit µg/insect		Percent inhibition (-)/ activation
			Untreated	Treated	
1.	Margosan-O	0.2500	2.6518	2.0320	(-)23.37%
2.	H-34	1.8000	1.7533	1.3404	(-)23.55%
3.	N6-b	56.0000	2.6518	1.7833	(-)33.32%
4.	DDVP	0.0086	0.1871	0.1706	(-) 9.67%

Table II Comparison of the RNA content in *Musca domestica* L., (PCSIR strain) after LD_{50}'s treatment of the respective tested compounds

S.No.	Compound	LD_{50}'s µg/fly	DNA content Mean unit µg/insect		Percent inhibition (-)/ activation
			Untreated	Treated	
1.	Margosan-O	0.2500	12.8334	10.3319	(-)19.49%
2.	H-34	1.8000	13.8599	8.9363	(-)35.52%
3.	N6-b	56.0000	12.8334	9.6229	(-)25.02%
4.	DDVP	0.0086	16.8067	21.6500	(-)28.82%

amount of RNA and DNA in comparison to the extracts of neem. In the present study DDVP decreased RNA and DNA content upto 9.67% and 28.82%, respectively.

The maximum inhibition of RNA was produced by N6-b (33.32%), whereas maximum inhibition of DNA content was 23.55% by H-34. In most of the cases inhibition percentile of RNA content was more than DNA content. This may be due to the fact in general total RNA content decreases rapidly after emergence of insect and total DNA decrease rapidly after emergence as reported by Chinzei and Tojo (1972) in *Bombyx mori*. That means a difference in RNA and DNA content naturally occurs in adult insect body, so possibly inhibition percentile is according to that difference. Inhibition of RNA and DNA contents by plant toxins or IGRs is further supported by the following reports. Philips and Loughton (1979) reported that actinomycine, diflubenzuron (IGR) and cyclophexamide inhibited RNA and protein synthesis in 4th instar nymph of *Locusta migratoria*. About 60% inhibition of RNA was found by these compounds. Shakoori *et al.*, (1985) reported the sublethal doses (200 ppm) as well as lethal doses (400 ppm) of a pyrethroid fenpropathrin against 6th instar larvae of *Tribolium castaneum* (Herbst.) did not change much of the RNA and DNA activity under laboratory condition, as in the present research with adult *M. domestica*. Saleem and Shakoori (1987) reported that DNA contents of *T. castaneum* remained unaltered when treated with sublethal doses (1 and 2 ppm) of permethrin, however, RNA content decreased to 16% and 28% at these concentration. Shakoori *et al.*, (1988) reported that at higher doses of fenpropathrin DNA and RNA contents decreased upto 20% and 21% respectively in 6th instar larvae of *T. castaneum*. In the present work lesser inhibition was noted after DDVP treatment. Low inhibition may be due to different mode of action. Thus higher inhibition of RNA and DNA in neem treated flies may be correlated with the fact that neem toxins are found to be growth retardant. So all the growth disruption, abnormal growth and finally death of the insect may be due to inhibition of nucleic acids. Consequently, there is inhibition of protein synthesis or impairment of RNA and DNA synthesis as reported by Philips and Loughton (1979). Shakoori *et al.*, (1994) reported elevation in DNA (21%) and RNA (28%) in Pak strain. While FSS-II strain showed 32% decrease in RNA activity and 44% increase in DNA contents after the treatment of sublethal dose (200 ppm) of Sumicidin super 6th instar larvae of *Tribolium castaneum*. Naqvi *et al.*, (1994) observed that the neem compound nimolicin inhibited 30.23% and 19.30% RNA and DNA contents. While solfac (pyrethroid) inhibited 17.70% and 15.63%, respectively. While in the present case decrease in RNA and DNA content was observed after neem compounds treatment (Margosan-O™, H-34, N6-b) and organophosphate (DDVP) treatment. The previous report support present findings however a slight difference between the results of both reports may be due to different chemical groups and structure of the toxins and method of application.

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